# SCHAEFFER, GRM AND THEIR INFLUENCE ON NES-TOOLS FOR MAX

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**ABSTRACT**

NES-Tools is a suite of audio signal processors emulating the GRM-Tools, but in the Max environment. This article provides the historical context for the creation of GRM-Tools, reaching back to the concepts developed by GRM's founder Pierre Schaeffer, providing a summary of technological developments at GRM, then describing the details of each of the NES-Tools. The adaptability of using the Max environment is demonstrated, and recent translations to Max for Live are noted.

## 1. INTRODUCTION

### 1.1. Aims

NES-Tools was inspired by the GRM-Tools suite of audio signal processors, created by the Groupe de Recherches Musicales (GRM) in Paris. The main aim for this project was to implement, within the Max[1] environment, the kind of audio signal processing tools found within GRM-Tools.

### 1.2. Background

In order to understand the unique collection that is GRM-Tools, it is helpful to examine the history behind this research group, and how the concepts that their researchers and composers developed were shaped by the technologies that underwent a period of rapid change, immediately after World War II.

The Groupe de Recherches Musicales (GRM) was set up by Pierre Schaeffer and the French Radio and Television service (RTF) in 1958 (Gayou 2007). Its predecessor was the Groupe de Recherche de Musique Concrète (GRMC, 1951–58). First we will examine the years before the formation of GRM, and the concepts Schaeffer developed as the years progressed (Palombini 1993; Gayou 2007). Then we will trace the origins and evolution of technologies used at GRM as articulated by Daniel Teruggi (Teruggi 2007).

## 2. SCHAEFFER AND THE PATH TO GRM

### 2.1. Before GRM

To understand both setup and the concepts espoused by GRM, we need to examine its gestation and the years leading up to its formation.

Schaeffer began experiments with radiophonic sound at the French Radio and Television service (RTF) dating back to 1942 (Gayou 2007). Beginning with the Studio d'Essai (1942–46), then the Club d'Essai (1946–60) and the Groupe de Recherche de Musique Concrète (GRMC, 1951–58), a unique approach to the composition of music using recorded sound was being developed.

### 2.2. Schaeffer Before 1942

Schaeffer was born in Nancy in 1910. He studied cello and went on to study engineering, specialising in telecommunications at the Ecole Polytechnique. He subsequently began his career as an engineer with the Radio Diffusion Française in 1936. In 1941 Schaeffer met George Gurdjieff, an Armenian guru who had travelled extensively but who had settled in Paris in 1922. Schaeffer was somewhat influenced by the teachings of Gurdjieff and Évelyne Gayou suggests that the notion of detachment from oneself and an opening up to the perception of other beings may have influenced Schaeffer's ideas about 'reduced listening', that is detaching sounds from their everyday meanings. (Gayou 2007, 204)

Working in groups was also a feature of Gurdjieff's teachings.

### 2.3. From 1942 (Beaune) to 1958 (GRM)

In the midst of World War II, and in German occupied France, Schaeffer collaborated with theatre director Jaques Copeau to organise training for practitioners of the emerging radiophonic arts: the Beaune training session. The trainees were chosen "for their pluri-disciplinary aptitude. They had to be actors, singers, musicians and technicians, all at the same time" (Gayou 2007, 205). The Beaune training session was "a model for Schaeffer, a successful symbiosis between art and craft, between research and creativity." (Gayou 2007, 205).

The Studio d'Essai (Trial Studio), was established in Paris on 12 November 1942. The Studio d'Essai "had to be both a laboratory of radiophonic art and a vocational training centre. Pierre Schaeffer at this time was working as the chief engineer at the radio, in charge of staff training." (Gayou 2007, 205). Not only did Schaeffer compose a radiophonic opera at this time, he also trained his team for secret activities in the Resistance, including the organisation of the radio broadcast calling for the liberation of Paris in August 1944. In quick succession,

[1] https://cycling74.com

after the liberation in 1945, he was made Director General of Radio, then fell out of favour and returned to Studio d'Essai, then was sent on a study tour of the USA and Canada to study radio there.

In 1946 the Club d'Essai followed on from the Studio d'Essai and was directed by poet Jean Tardieu. It was the start of an avant-garde period as the whole art world was more intent on exploring abstraction in the period after the war. Schaeffer was variously involved with the Centre d'Études Radiophonique and the Ministry of Foreign Affairs. In between foreign trips, he would return to the Club d'Essai to continue his experiments and develop his conceptual thinking. The period of research in the late 1940s by Schaeffer and his colleagues marked "the passage from realistic sound effects adapted to radiophonic theatre to a musical abstraction" (Gayou 2007, 205).

Schaeffer experimented with sound effects at the French Radio facility and this research into noise resulted in six "noise studies" (Palombini 1993, 14). The relationship between the abstract and the concrete and the process by which realisation of music is carried out were also issues explored by Schaeffer.

The very first appearance of the term 'musique concrète'[2] was at the presentation of Pierre Schaeffer's 'Concert de bruits' on Sunday 20 June 1948, at the Club d'Essai in Paris. As quoted by Gayou, Pierre Schaeffer made the notable declaration:

> To reach music, one can take an inverse path, which is to set out from sound data instead of notation, and from a former composition instead of a traditional execution. If the term did not seem so pretentious, we would call these attempts 'attempts at Musique Concrète', in order to better define their general character and because it is not so much a question of noise but of a method of musical composition. (Gayou 2007, 203)

Also, there was mounting interest created by the concert de bruits and the broadcast of a number of the noise studies by French Radio on Tuesday 5th October, 1948. This interest eventually led to "the publication of the article *Introduction a la musique concrète* (Schaeffer 1950), in which he felt reassured enough to replace the phrase 'research into noises' by the more ambitious 'musique concrète'." (Palombini 1993, 16)

Schaeffer's next work, in 1949, was the *Suite no. 14* (Suite for 14 Instruments). With the help of orchestrator Jean-Michel Damasse, Schaeffer produced a score for 14 orchestral instruments, which was then recorded. The recording was then "decomposed, shortened, magnified, dissected, inverted, exploded, pulverized" (Schaeffer 1952, 39). The theory being that if noises could provide material for musical construction, then less "arid" material should produce better results. However, what emerged was full of contradictions, and Schaeffer concluded that musique concrète produced powerful techniques, but that it lacked a theoretical grounding.

In 1949, Schaeffer also recruited young percussionist Pierre Henry to assist by performing on various sound objects in recording sessions. Their collaboration resulted in the *Symphonie pour un homme seul* (Symphony for a man alone) in 1949-1950. It was performed at the Salle de l'Empire in Paris on 6 July 1951.

In 1951, Schaeffer also lectured at the Darmstadt festival, where the Symphonie was enthusiastically received. His ideas were likewise warmly welcomed by German technicians.

Within the structure of the French national radio, Schaeffer formed the Groupe de Researches de Musique Concrète (GRMC) in 1951. GRMC was organised according to Gurdjieff's principles and experienced individuals trained the new composers. It attracted curious musicians and some of the "great names of the day followed one another: Andre Hodeir, Pierre Boulez, Olivier Messiaen, Michel Philippot, Monique Rollin, Karlheinz Stockhausen, Jean Barraque, Darius Milhaud, Edgar Varèse, Henri Sauguet and Roman Haubenstock-Ramati." (Gayou 2007, 206)

The First International Decade of Experimental Music was organized by the Groupe de Recherches de Musique Concrète in Paris in June 1953. It was an attempt to unite the contemporary music world under the banner of experimental music, but whose experimental music? Given that there was even a schism within GRMC between composers like Pierre Boulez, who thought that technology was merely a means for the improvement of the serial tradition and Schaeffer and Henry, who were trying to come to terms with some sort of conceptual framework for the organisation of musique concrète (as opposed to being merely a method of transforming sounds). Schaeffer was later to publish his ideas in an article titled "Vers une musique experimentale" (Schaeffer 1957).

Some years earlier (1951) Schaeffer and Henry had embarked on the creation of a concrete opera *Orphée* (1951-1953). The combination of song and tape music was "considered a scandal at its performance at Donaueschingen in (October) 1953, as if it had been a crime of high treason against the avant garde".[3]

After the Decade festival and the controversy of the *Orphée 53* performance, Schaeffer distanced himself from the GRMC to direct the foundation and management of Radiodiffusion de la France d'Outremer – the international arm of the French broadcaster.

As Gayou notes, there was also some resentment, by Schaeffer, of the success being enjoyed by Pierre Henry and Philippe Arthuys. So much so that, when Schaeffer

[2] Gayou (Gayou 2007, 203) quotes Schaeffer from his book Pierre Schaeffer, 1951, *A la recherche d'une musique concrète*. In the first part of this book, Schaeffer recounts its invention in the form of a diary

[3] As described by Michel Chion in the notes (pp. 55-56) to Schaeffer's compilation CD set, see Schaeffer (1998)

returned to the GRMC in 1957, he forced them to resign and he restructured the research group into the Groupe de Recherches Musicales (GRM). "A few months later, Schaeffer recruited his new collaborators: Luc Ferrari, Iannis Xenakis, Bernard Parmegiani, Mireille Chamass-Kyrou, François-Bernard Mâche. Later arrived Ivo Malec, Philippe Carson, Romuald Vandelle, Edgardo Canton and François Bayle." (Gayou 2007, 207)

With the new GRM, Schaeffer had withdrawn the term musique concrète, with its aesthetic connotations, and he started defining his work as music research. He returned to the studio and composed a new set of studies. In 1960 he established a broader research department with himself as head. It included the GRM along with the Groupe de Recherches Image GRI, the Groupe de Recherches Technologiques GRT and the Groupe de Recherches Langage which became the Groupe d'Etudes Critiques. In 1966, Schaeffer published his treatise on musical objects, the *Traité des Objets Musicaux* (TOM) – the summation of all his work. "The Solfége de l'Objet Sonore came out in the following year and consisted of three LPs of sonic examples that served to illustrate the *Traité*." (Gayou 2007, 207-208)

Schaeffer left the running of GRM to François Bayle while he concentrated on Le Service de la Recherche until its replacement by l'Institut National de l'Audiovisuel in 1975, when Schaeffer was forced to retire at the age of 65.

In summary, we can track Schaeffer's ideas and activities from radiophonic art to noise studies, musique concrète and experimental music. From the detached sound object to a new interpretation of the note, instrument, and context for expression (the concert). The mantra of "listen then do" was used throughout the history of both the GRMC and GRM. A comprehensive solfège of the sound object was combined with a theoretic framework expressed in the TOM publication. With each idea, and at each stage, new technologies were embraced as fresh enablers. The notable exception, for Schaeffer, was the digital computer, which he regarded as an obstacle between the composer and music making. With this historical, conceptual and organisational background in place, we will take up the topic of technology in the next section.

## 3. GRM TECHNOLOGIES

### 3.1. Musique Concrète Concepts

In the first decade of GRM there are concrete sounds (re-modelled[4]), instrumental sounds (re-modelled), and a short time later, electronic sounds being used. The introduction of electronic sounds coincided with a relaxation of the initial strong rules surrounding early musique concrète to "…leave place for a more free and personal way of using sounds, for which every composer designs his personal contour" (Teruggi 2007, 214)

Whatever the source, the notion of structuring of various sound objects must be achieved through listening and derived by abstracting formal structures from the source materials themselves, as opposed to superimposing an 'a priori' musical system on the materials. This was the fundamental difference between Schaeffer's approach and it put him at odds with the method employed by the serial composers, like Pierre Boulez, who were a dominant force in the 1950s.

With the Schaefferian approach, there was a need for new tools to modify and manipulate sounds to heighten abstraction: "…with the objective of producing sounds that will be perceived primarily as forms and structure and less as anecdotes or language references. Instrumental sounds can be combined; however, concrète sounds have to be modelled before being combined." (Teruggi 2007, 214) It is also important to realise that any tools that are used will leave their own traces that can be heard by the ear. But what were these tools?

### 3.2. Technological Periods

In his article in the Computer Music Journal, Teruggi breaks up the history of technology in GRM into four main periods, the first of which pre-dates the GRM (Teruggi 2007, 215):

- Mechanical period: 1948 - 1960.
- Electronic period: 1960 - 1970.
- Digital mainframe period: 1970 - 1990.
- Personal computer period: 1990 – present day.

### 3.3. The Mechanical Period

Around 1948, Pierre Schaeffer's studio comprised a shellac record recorder, shellac record players, a mixing desk, mechanical reverberation unit, several kinds of filters, and microphones.

This equipment afforded the following musical possibilities: sound transposition by playing recordings at various speeds; playing forwards and reverse; sound looping by creating a continuous groove in a shellac record; sample extraction by selecting a particular part of a record; filtering to change the timbre; mixing sounds by recording from the mixer; and creative microphone techniques to capture different aspects of a sound making object.

Tape recorders arrived in 1949, but were initially found to be less reliable than shellac players: "…to the point that the *Symphonie pour un homme seul*, which was composed in 1950–51, was mainly composed with records, even if the tape recorder was available" (Teruggi 2007, 216).

[4] "Modelling" a sound is the process by which filtering, transposition and microphone placement erase the source reference and retain essential information for musical use.

Around 1950, Schaeffer elicited the help of technician Jacques Poullin to create new machines that were derived from the tape recorder. Two types of phonogène (sound generator) were created in 1953 to allow pitch transposition of recorded material. The chromatic phonogène had a one octave keyboard and 12 reading heads associated with capstans of different sizes. The 12 possible speeds permitted transposition by 12 different pitches. The sliding phonogène made continuous speed variation, and therefore glissandi, possible.

### 3.4. The Electronic Period

The widespread availability of the transistor in the late 1950s led to a lot of development in electronics. The modular synthesizer was one of such inventions. Although recording sounds was the focus at GRM, Pierre Henry had started using oscillators to produce sounds as early as 1955. But the emerging synthesisers meant parameter control, which was against Schaeffer's philosophy of "making through listening" (Teruggi 2007, 219).

However, a project for the development of the Coupigny synthesiser was initiated. It was created under Schaeffer's influence where "parameters would be globally controlled without actually permitting a very precise definition of their values" (Teruggi 2007, 220). Thus it functioned more as a sound event generator rather than a precision musical instrument.

The Coupigny synthesiser included the major components of a modular synthesiser that could be easily interconnected: oscillators, noise-generators, filters, ring-modulators, with facilities for frequency modulation, amplitude modulation, and external source modulation. It had a complex envelope generator, but no keyboard was attached. Thus it could produce extremely complex sounds, but it was less suited to producing precise frequencies or triggering sounds in a sophisticated way. "The typo-morphological concepts of Schaeffer were clearly applied here." (Teruggi 2007, 220)

The Coupigny synthesiser breathed new life into the GRM composers and many GRM masterworks were composer from 1969 to 1975, including the classic *De natura sonorum* by Bernard Parmegiani (1974-75), which used concrete sounds, instrument sounds, and synthesiser sounds in an interwoven fabric that was carefully crafted.

### 3.5. The Mainframe Computer Period

Jean-Claude Risset had been collaborating with Max Mathews at the Bell Laboratories since 1964. Risset brought some of the first works calculated by main frame computer to France, along with the software synthesis language Music V. A group of young GRM researchers were interested but Schaeffer was sceptical about computers since they required a pre-conceptualisation of a musical work. Teruggi relates a critical turning point for Schaeffer and GRM in 1970:

> An international conference on computer music took place in 1970 in Stockholm, fostered by Unesco. Many of the leading researchers in the digital domain were present at the meeting, including Mathews, Risset, Zinovieff and Schaeffer as a kind of historical reference. During this meeting Schaeffer gave a speech and participated in a roundtable discussion, where he was extremely critical towards computer-generated sounds and their eventual use in electroacoustic music. Participants at the conference were quite shocked by the conservative attitude of one of the inventors of technological music and the consequences were a certain isolation of the GRM from the digital actors. (Teruggi 2007, 220-221)

The GRM technical team was disappointed by this attitude and decided to explore two avenues simultaneously. The first team saw a way forward with the development of possible real-time synthesis solutions, which would still enable a composer to interact with sound materials directly. The second wanted to understand how the Music V software could be applied to musique concrète.

Real-time synthesis was very difficult using computers at this time, and there were several attempts. One project started as a collaboration between Knut Wiggen from EMS in Stockholm and the GRM. Christened Syntom from SYN(thèse) + T(raité) des O(bjets) M(usicaux) (synthesis based on the treatise of musical objects). It took as its starting point the concepts expressed by Schaeffer in the *Traité.* Attempting to create typologies and morphologies with buttons would prove too difficult, and the project failed.

Meanwhile, the team of Pierre-Alain Jaffrennou, Benedict Mailliard experimented with Music V towards a GRM-based way of composing. Then in 1974, a young engineer by the name of Jean-François Allouis arrived. The invention of smaller, faster chips meant that faster parameter control was becoming feasible and the notion of hybrid synthesis controlled by such chips was making real-time systems possible. The Syter Project started in 1975 with Syter signifying firstly 'real-time synthesis' then later 'real-time system'. The first prototype was used in 1977.

In 1978, GRM acquired its own mainframe computer, running Music V. A PDP 11/60 mini-computer was also installed with its own digital-to-analog converters. The team was located in Studio 123 at Radio France, and developed models for action on sound in software.

### 3.6. Transition to Real-time and the SYTER Project

Slowly the work of the Studio 123 team and Allouis' real-time work came together and the first fully operational Syter system was finished in 1984.

The Syter system was comprised of modules such as oscillators, envelopes, delays, harmonisers, and noise-generators. These modules were able to be inter-connected with each other and with an innovative set of sound processors. Many of these were derived from the models developed by the Studio 123 software synthesis group. Teruggi lists some of these processing instruments and includes their corresponding name in the GRM Tools set. (Teruggi 2007, 226) It also had a graphic interface

controlled through a mouse, thus permitting real-time control of the parameter variation and a MIDI interface.

### 3.7. Personal Computer Period - GRM Tools

Hugues Vinet, replaced Jean-François Allouis in 1988, and started experimenting using the new Apple Macintosh with the Digidesign Sound Designer card, which had become available then. Using this system, the first version of GRM Tools appeared in 1991. Although constrained by a reduction in processing power, and therefore a step back from the Syter system, it provided broader access to the software tools.

Emmanuel Favreau continued the development in 1994 by adapting GRM Tools to new environments. Teruggi notes:

> A first series of eight algorithms was launched called Classical bundle, which was mainly an adaptation of the Studio 123 and Syter models to the new environments, with improved graphical controls and performances, and sometimes with new possibilities. (Teruggi 2007, 226)

GRM Tools ST followed, allowing spectral processing and analysis-synthesis methodologies.

### 3.8. Summary of GRM Technology Concepts

At GRM, a set of operating categories had evolved over the years and were given the name 'morpho-concepts' by Hugues Vinet and François Bayle in the early 1990s. In a way, these concepts unite much of the technology from the 1950s onward. Teruggi lists[5] them as: Sound isolation and observation; Sound editing; Sound addition; plus the following "modifications": Dynamic; Speed; Time; Spectral; Density; Order of events; and Space. (Teruggi 2007, 229)

Teruggi also cites three trends that have influenced every research and development team at GRM. In other words, these are the basic design principles underpinning all of the technical developments at GRM (Teruggi 2007, 229):

1. Tools are made to be controlled through listening and during listening. The basic concept issued from the 'making' and 'listening' duality by Pierre Schaeffer.
2. Tools should be easy to grasp and manipulate.
3. The user should be independent and not rely on an external operator.

## 4. NES-TOOLS

### 4.1. NES-Tools overview

Inspired by the GRM Tools suite of audio processors, this project set out to create a set of processing tools using Cycling 74's Max environment. The idea was to just use only the standard objects that come with Max 7/8, however many of the tools I wanted to create had already been implemented in one form or another in the standard distribution files and examples. What I have done in this project is to adapt and add to the examples created by others, and then assemble these tools into their own suite. The NES-Tools suite is summarized in Table 1, with the corresponding GRM Tools Classic equivalent in italics.

| | |
|---|---|
| NES-Biquad~ *Bandpass* | Implements a two-pole two-zero filter (for 2 channels) |
| NES-Brassage~ *Shuffling* | Granulate a sound loaded into the device itself |
| NES-CombFilter5~ *Comb Filter* | Implements 5 Comb filters |
| NES-DelayTaps~ *Delays* | Up to 32 delay taps |
| NES-DopplerPan *Doppler* | Simulate a Doppler effect |
| NES-PitchMulti~ *PitchAccum* | Multiple pitch shifters with vibrato & feedback |
| NES-Reson~ *Reson* | Up to 32 simultaneous resonators |

**Table 1**. NES-Tools suite of transformations.

### 4.2. NES-Tools Overview Patcher

To use the NES-Tools suite, the file 01-NES-Tools-Overview.maxpat provides a quick way for the user to try each tool by running the patcher in Max and clicking on the relevant title (See Fig. 1). Make sure you close a patch before opening another one, to avoid any confusing conflicts.

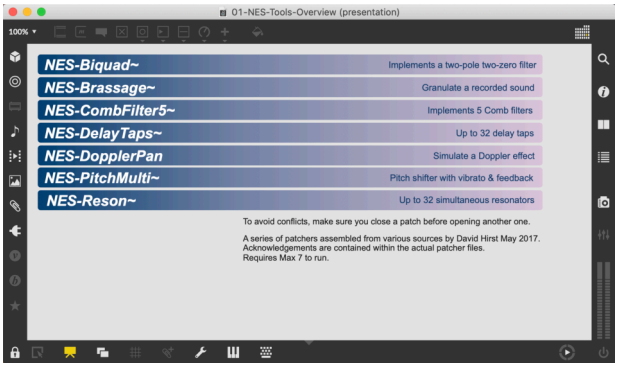



**Figure 1**. The NES-Tools Overview patcher[6].

### 4.3. NES-Tools details

Except for the special case of NES-Brassage~, each of the NES-Tools is set out in the same way. Figure 2 shows NES-Biquad~ and its structure is set out from top to

[5] Only their headings are listed here. See the Teruggi (2007) article for a full explanation of each

[6] Note that the screen shots in this article depict the general layout only. For detailed views, download the NES-Tools suite from https://davidhirst.me/software/ and run the patchers using Max 8 from https://cycling74.com

bottom. At the top is a section where you load a sound file. In the middle section there are the controls for the relevant processor. At the bottom of the patcher is a section that enables the user to adjust the playback gain and to record a sound file of the processed sound. The user can then quickly load and audition the recorded sound file.

Here are more specific details for each tool.

**NES-Biquad~**

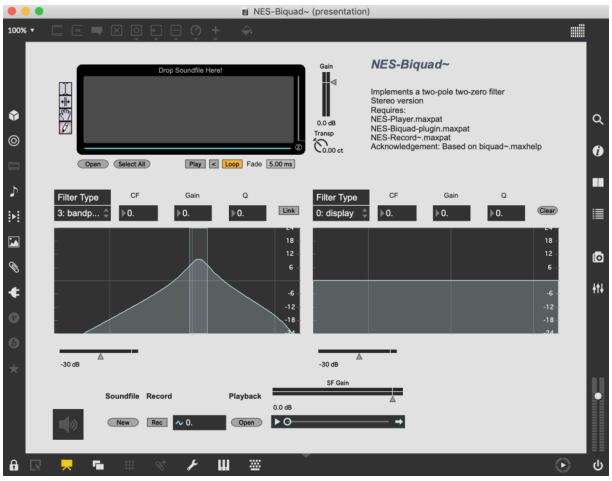



**Figure 2**. NES-Biquad~

NES-Biquad~ (Fig. 2) is based on biquad~.maxhelp and implements a two-pole two-zero filter (for 2 channels). Change the filter type, frequency, gain and Q for two channels independently, or use the 'Link' switch to allow the left controls to apply to the right channel simultaneously.

**NES-Brassage~**

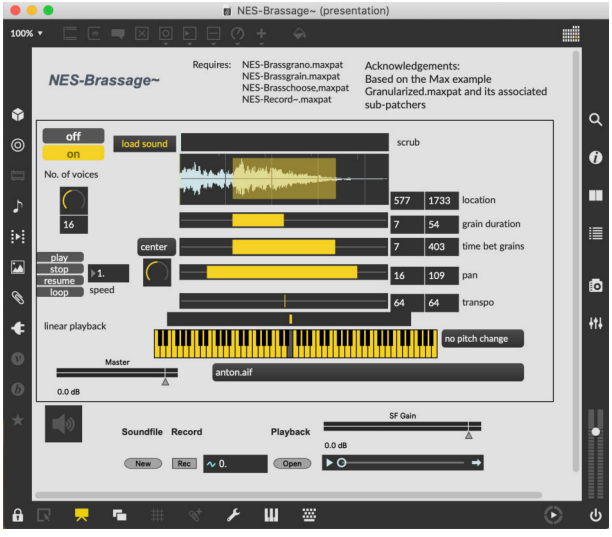



**Figure 3**. NES-Brassage~

NES-Brassage~ (Fig. 3) granulates a recorded sound, and is based on the Max example Granularized.maxpat and its associated patchers.

The user loads a sound file and clicks 'on' to start granulation, then either selects a section in the waveform view, or presses play or loop to provide the source for granulation. The granulation parameters can then be varied.

Controls are for: grain location within the source file; grain duration; time between grains; pan position; and pitch transposition. A range can also be set for all of these parameters, within which random values are constrained.

Other controls are the number of granular "voices" or simultaneous grains; the speed of playback of the source file; scrubbing the source file; and overall gain. The resultant sound can be recorded.

**NES-CombFilter5~**




**Figure 4**. NES-CombFilter5~

NES-CombFilter5~ (Fig. 4) implements 5 Comb filters, and is based on based on the CombFilterMulti Max for Live plugin. There are controls for pitch; detune; and feedback for each of 5 bands, plus input gain; feed-forward; transposition; tremolo; dry/wet; and an output gain. A frequency display completes the tool design.

**NES-DelayTaps~**

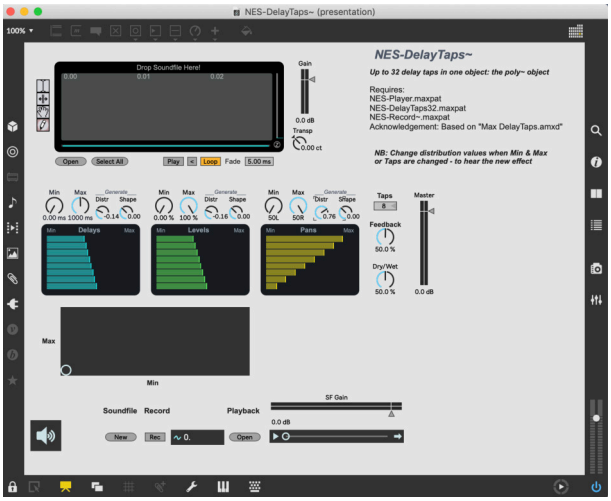



**Figure 5**. NES-DelayTaps~

NES-DelayTaps~ (Fig. 5) implements up to 32 delay taps. The tool is based on a modified version of Max DelayTaps.amxd, distributed with Max 7. There are controls for delays; levels; and pans for up to 32 separate

delay lines. Each of these parameters has its own controls for min/max, distribution and shape and an editable display of their values. The X-Y controller allows you to change the min/max delay amounts simultaneously. Also vary the number of taps, overall feedback, dry/wet, and master gain.

**NES-DopplerPan**

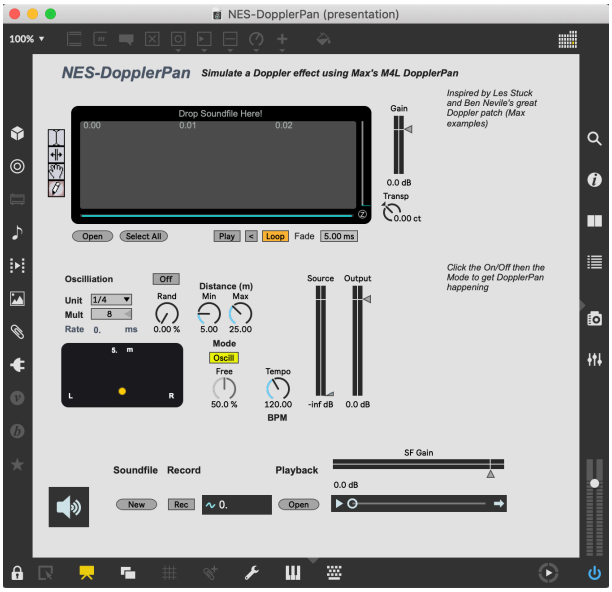



**Figure 6**. NES-DopplerPan

NES-DopplerPan (Fig. 6) simulates a Doppler effect and is a modification of Max for Live's DopplerPan, which was inspired by Les Stuck and Ben Nevile's Doppler patch (from the Max examples). The user clicks the On/Off, then the Mode, sets the min/max distance, decides on the degree of randomness, etc. The pan oscillation can be synchronised with a tempo.

**NES-PitchMulti~**

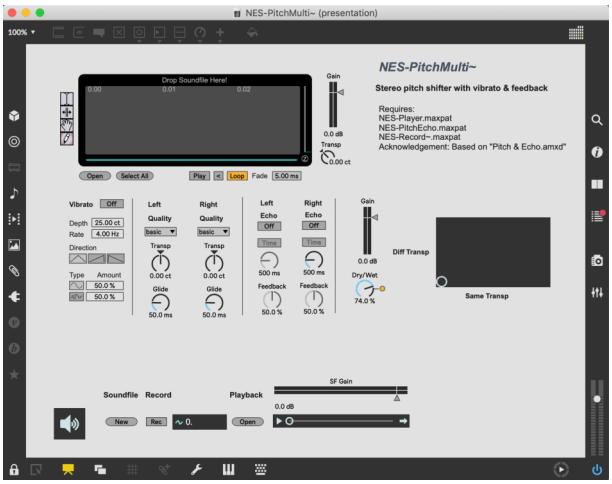



**Figure 7**. NES-PitchMulti~

NES-PitchMulti~ (Fig. 7) pitch shifts with vibrato and feedback, and is derived from Pitch & Echo.amxd, distributed with Max 7. The X-Y controller allows you to change the transposition amounts for each channel simultaneously.

**NES-Reson~**

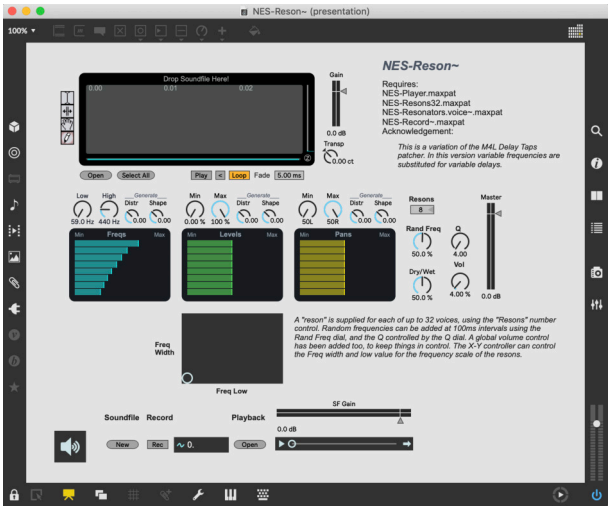



**Figure 8**. NES-Reson~

NES-Reson~ (Fig. 8) implements up to 32 simultaneous resonators. It is a variation of the Max for Live Delay Taps patcher where variable frequencies are substituted for variable delays. Random frequencies can be added to the resonators, and the Q is controlled by its own dial. A global volume dial has been included to keep the Q under control, plus there is a dry/wet dial and a master fader. The X-Y controller allows the user to simultaneously change the lower frequency value and the difference between this lower value and the upper value for the range of resonances.

### 4.4. NES-Tools Abstractions and Extensions

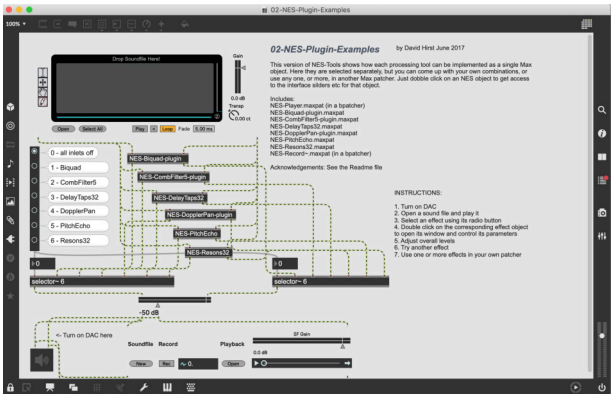



**Figure 9**. 02-NES-Plugin-Examples.maxpat

All of the tools (except NES-Brassage~) implement their processing as a separate patcher, and can therefore become their own object in Max. This is known as "abstraction" in Max. The user can create her/his own master patcher that could string a series of processes together in the one patcher. 02-NES-Plugin-Examples.maxpat provides one example of how this can be done (Fig. 9). In the example the user can open a sound file and play it; select an effect using its radio button; double click on the corresponding effect object to open its UI window and control its parameters; adjust overall levels; try another effect; and use one or more effects in their own patcher.

### 4.5. NES-Tools - Max for Live Versions

In a recent development, NES-Tools has been written as a suite of plugin devices for Ableton Live, mostly as "Max Audio Effects".

NES-Brassage02.amxd (Fig. 10) is implemented as a Live "Max Instrument" and has been specially programmed to allow for multiple instances in Ableton Live, as have all the above devices.

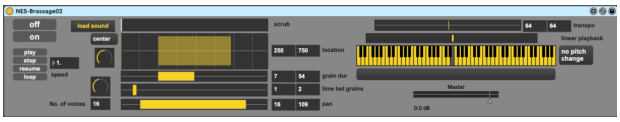


**Figure 10**. NES-Brassage02.amxd

## 5. EXAMPLE OF COMPOSITIONAL USE

As a final test, the NES-Tools have been used, along with the NES-Spectrale suite, MUBU and spindrift~ to create the 15 minute work *The Portuguese Suite*[7], performed in its eight channel version at ACMC 2019.

Note that the resultant final sounds within the work are a combination of many processes, but here are some selected example passages from the recording that the reader can find via the Soundcloud link (See Table 2 and the footnotes).

| | |
|---|---|
| 00:00 – 00:58 | Uses NES-Reson~ to elongate the bell sounds |
| 00:56 – 03:02 | NES-PitchMulti~ combined with interpolation of bells |
| 01:58 – 03:56 | FFT screening FX of bells from NES-Spectrale |
| 03:55 – 05:12 | Transformed clapping from NES-Spectrale |
| 05:14 – 07:48 | Uses the Playtwo effect from NES-Spectral |
| 07:50 – 10:45 | Uses the CataRT & Mosaic FX from IRCAM's MUBU suite[8] |
| 09:48 – 14:16 | Combined FX of NES-Spectrale variable playback, NES-Biquad~ & other filtering, & the spindrift~ object by Michael Norris[9] |

**Table 2**. Selected examples of tool use in *The Portuguese Suite*.

[7] The Portuguese Suite can be listened to at https://soundcloud.com/david-hirst-1/portuguese-suite

[8] MUBU is available from the IRCAM Forum https://forum.ircam.fr/projects/detail/mubu/

## 6. CONCLUSION AND FURTHER DEVELOPMENTS

The virtue of NES-Tools is that the suite provides a GRM-Tools-like set of audio processors within the Max environment. That means the patchers can be used, as they are, with the free Max version in "non-saving mode", or they can be edited and extended by the user with a paid copy of Max. NES-Tools written as Max for Live devices are also free from the author's website.

Future developments will explore the use of presets and the recording and storage dynamic controls. Custom configurations of NES-Tools can also be used as live performance instruments, encouraging improvisational uses.

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[9] Spindrift~by Michael Norris is available here: http://www.michaelnorris.info/software/spindrift